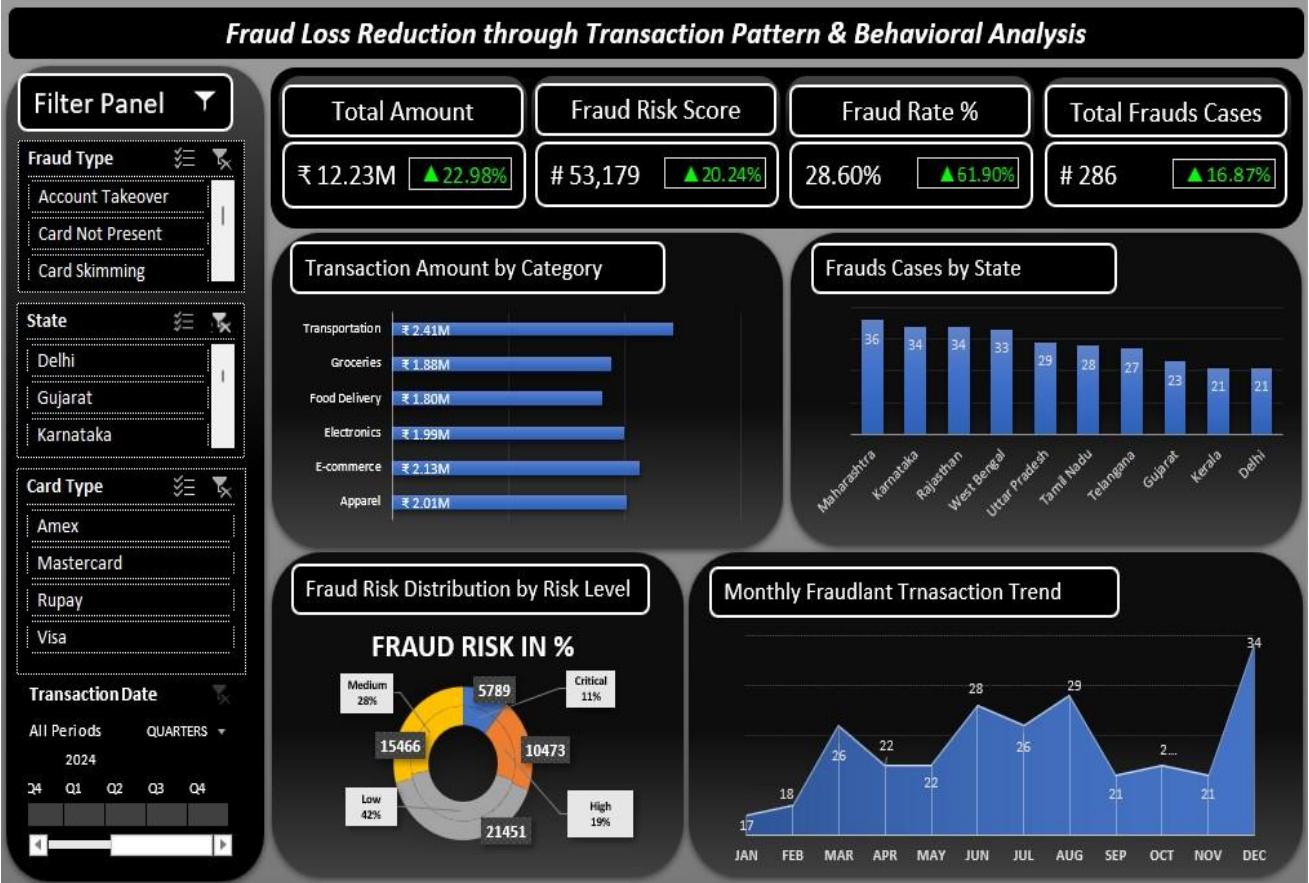


Fraud Loss Reduction through Transaction Pattern & Behaviour Analysis

This professional report presents a comprehensive fraud transaction analysis based on transaction behavior, fraud patterns, category-level risk exposure, and regional fraud trends. The analysis aims to identify major fraud contributors and recommend actionable strategies to reduce financial losses and strengthen transaction security.

Dashboard Overview



The dashboard provides an overview of fraud activity across transaction categories, states, card types, and fraud risk levels. KPI cards highlight total fraud amount, fraud risk score, fraud rate, and total fraud cases, helping stakeholders quickly identify fraud exposure and operational risk.

KPI Summary

Metric	Value
Total Fraud Amount	₹ 12.23M
Fraud Risk Score	53,179
Fraud Rate	28.60%
Total Fraud Cases	286

Key Insights

Fraud Loss Reduction through Transaction Pattern & Behavioral Analysis - KEY INSIGHTS

1. High Overall Fraud Exposure:

- ★ Total fraud amount reached ₹12.23M, indicating significant financial loss due to fraudulent transactions.
- ★ The fraud rate stands at 28.60%, showing that nearly one-third of flagged transactions are fraudulent.

2. Transportation Category Has Highest Fraud Amount:

- ★ Transportation recorded the highest fraudulent transaction amount at approximately ₹2.41M.
- ★ E-commerce and Apparel categories also showed elevated fraud activity, indicating high-risk digital and card-based transactions.

3. Maharastra Leads in Fraud Cases:

- ★ Maharashtra reported the highest number of fraud cases among all states.
- ★ Karnataka and Rajasthan also showed consistently high fraud activity, suggesting regional fraud concentration.

4. Fraud Activity Increased Towards Year-End:

- ★ Monthly fraud trend analysis shows a gradual increase in fraud cases during the second half of the year.
- ★ December recorded the highest fraud count, indicating possible seasonal fraud spikes during high transaction periods.

5. Medium and Low Risk Transactions Dominate:

- ★ Medium-risk and low-risk transactions contribute the largest portion of overall fraud distribution. This suggests fraudsters may intentionally keep transaction behavior below high-risk detection thresholds.

The analysis indicates that transportation, e-commerce, and apparel categories experienced the highest fraud exposure. Maharashtra recorded the largest number of fraud cases, while fraud activity increased significantly during the second half of the year, especially in December. Medium- and low-risk transactions contributed heavily to the fraud distribution, suggesting the need for more advanced fraud detection systems beyond traditional high-risk flagging.

Business Recommendations

Fraud Loss Reduction through Transaction Pattern & Behavioral Analysis - BUSINESS RECOMMENDATIONS

1. Increase Monitoring in High Fraud Categories:

- ★ Pay extra attention to:
 - Transportation
 - E-commerce
 - Apparel
- ★ Add stronger checking systems for transactions in these categories.

2. Deploy State-Level Fraud Control Strategies:

- ★ Increase fraud monitoring and investigation efforts in:
 - Maharashtra
 - Karnataka
 - Rajasthan
- ★ Introduce region-based fraud scoring models.

3. Enhance Transaction Verification:

- ★ Introduce additional authentication measures such as:
 - Multi-factor authentication (MFA)
 - OTP validation
 - Biometric verification for high-value transactions

To reduce fraud losses, the organization should strengthen monitoring in high-risk transaction categories and states, implement multi-factor authentication and OTP verification, improve fraud scoring systems, and increase monitoring during seasonal transaction spikes. Real-time transaction alerts and region-based fraud detection models can further help prevent suspicious activities and improve customer trust.

Conclusion

This analysis highlights critical fraud trends affecting transaction security and operational performance. By implementing proactive monitoring strategies, advanced fraud verification systems, and targeted fraud prevention initiatives, organizations can significantly reduce fraud losses and strengthen customer confidence.